
```

load('lin_sys.mat', 'sys');

Ts = 0.05; % Sampling time (in seconds)
sys_dt = c2d(sys, Ts, 'zoh'); % Discrete-time state-space model

% A matrix of augmented system
% the eye(3, 3) is tracking only the state of (x, y, z)
NA = [
    eye(3, 3), eye(3, 12);
    zeros(12, 3), sys_dt.A
]
NB = [zeros(3, 4); sys_dt.B]

CM = ctrb(NA, NB);
assert(rank(CM) == 15)

Q = diag([0.2, 0.2, 0.5, 2, 2, 20, 1, 1, 1, 2, 2, 0.1, 0.1, 0.1, 0.1])
R = eye(4)

[K, S, e] = dlqr(NA, NB, Q, R, zeros(15, 4))

Ki = K(:,1:3)
Ks = K(:,4:end)

NA =

Columns 1 through 7

    1.0000         0         0    1.0000         0         0         0
         0    1.0000         0         0    1.0000         0         0
         0         0    1.0000         0         0    1.0000         0
         0         0         0    1.0000         0         0    0.0494
         0         0         0         0    1.0000         0         0
         0         0         0         0         0    1.0000         0
         0         0         0         0         0    1.0000    0.9753
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0

Columns 8 through 14

         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0         0         0         0         0
         0         0         0    0.0122         0         0    0.0002

```

0.0494	0	-0.0122	0	0	-0.0002	0
0	0.0494	0	0	0	0	0
0	0	0	0.4844	0	0	0.0122
0.9753	0	-0.4844	0	0	-0.0122	0
0	0.9753	0	0	0	0	0
0	0	1.0000	0	0	0.0500	0
0	0	0	1.0000	0	0	0.0500
0	0	0	0	1.0000	0	0
0	0	0	0	0	1.0000	0
0	0	0	0	0	0	1.0000
0	0	0	0	0	0	0

Column 15

0
0
0
0
0
0
0
0
0
0
0
0
0.0500
0
0
1.0000

NB =

0	0	0	0
0	0	0	0
0	0	0	0
0	0.0000	0	-0.0000
-0.0000	0	0.0000	0
0.0001	0.0001	0.0001	0.0001
0	0.0003	0	-0.0003
-0.0003	0	0.0003	0
0.0030	0.0030	0.0030	0.0030
0.0019	0	-0.0019	0
0	0.0019	0	-0.0019
0.0001	-0.0001	0.0001	-0.0001
0.0750	0	-0.0750	0
0	0.0750	0	-0.0750
0.0050	-0.0050	0.0050	-0.0050

Q =

Columns 1 through 7

0.2000	0	0	0	0	0	0
0	0.2000	0	0	0	0	0
0	0	0.5000	0	0	0	0
0	0	0	2.0000	0	0	0
0	0	0	0	2.0000	0	0
0	0	0	0	0	20.0000	0
0	0	0	0	0	0	1.0000
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

Columns 8 through 14

0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
1.0000	0	0	0	0	0	0
0	1.0000	0	0	0	0	0
0	0	2.0000	0	0	0	0
0	0	0	2.0000	0	0	0
0	0	0	0	0.1000	0	0
0	0	0	0	0	0.1000	0
0	0	0	0	0	0	0.1000
0	0	0	0	0	0	0

Column 15

0
0
0
0
0
0
0
0
0
0
0
0
0
0
0
0.1000

R =

1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

K =

Columns 1 through 7

0.0000	-0.2520	0.3364	0.0000	-6.2934	12.0470	0.0000
0.2520	-0.0000	0.3364	6.2934	-0.0000	12.0470	3.1129
-0.0000	0.2520	0.3364	-0.0000	6.2934	12.0470	-0.0000
-0.2520	0.0000	0.3364	-6.2934	-0.0000	12.0470	-3.1129

Columns 8 through 14

-3.1129	8.0631	11.6872	-0.0000	0.1567	2.7120	-0.0000
-0.0000	8.0631	0.0000	11.6872	-0.1567	0.0000	2.7120
3.1129	8.0631	-11.6872	-0.0000	0.1567	-2.7120	0.0000
-0.0000	8.0631	0.0000	-11.6872	-0.1567	0.0000	-2.7120

Column 15

0.8989
-0.8989
0.8989
-0.8989

S =

1.0e+03 *

Columns 1 through 7

0.0050	0.0000	-0.0000	0.0589	0.0000	-0.0000	0.0177
0.0000	0.0050	-0.0000	0.0000	0.0589	-0.0000	0.0000
-0.0000	-0.0000	0.0179	-0.0000	-0.0000	0.3016	-0.0000
0.0589	0.0000	-0.0000	1.0184	0.0000	-0.0000	0.3539
0.0000	0.0589	-0.0000	0.0000	1.0184	-0.0000	0.0000
-0.0000	-0.0000	0.3016	-0.0000	-0.0000	8.1404	-0.0000
0.0177	0.0000	-0.0000	0.3539	0.0000	-0.0000	0.1448
0.0000	0.0177	-0.0000	0.0000	0.3539	-0.0000	0.0000
-0.0000	-0.0000	0.1178	-0.0000	-0.0000	4.1603	-0.0000
-0.0000	-0.0385	0.0000	-0.0000	-0.8565	0.0000	-0.0000
0.0385	0.0000	-0.0000	0.8565	0.0000	-0.0000	0.3809
0.0000	0.0000	-0.0000	0.0000	0.0000	0.0000	0.0000
-0.0000	-0.0043	0.0000	-0.0000	-0.1039	0.0000	-0.0000
0.0043	0.0000	-0.0000	0.1039	0.0000	-0.0000	0.0505
0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Columns 8 through 14

0.0000	-0.0000	-0.0000	0.0385	0.0000	-0.0000	0.0043
0.0177	-0.0000	-0.0385	0.0000	0.0000	-0.0043	0.0000
-0.0000	0.1178	0.0000	-0.0000	-0.0000	0.0000	-0.0000
0.0000	-0.0000	-0.0000	0.8565	0.0000	-0.0000	0.1039
0.3539	-0.0000	-0.8565	0.0000	0.0000	-0.1039	0.0000
-0.0000	4.1603	0.0000	-0.0000	0.0000	0.0000	-0.0000
0.0000	-0.0000	-0.0000	0.3809	0.0000	-0.0000	0.0505
0.1448	-0.0000	-0.3809	0.0000	0.0000	-0.0505	0.0000
-0.0000	2.7559	0.0000	-0.0000	0.0000	0.0000	-0.0000
-0.3809	0.0000	1.1721	-0.0000	-0.0000	0.1836	-0.0000
0.0000	-0.0000	-0.0000	1.1721	0.0000	-0.0000	0.1836
0.0000	0.0000	-0.0000	0.0000	0.0115	-0.0000	0.0000
-0.0505	0.0000	0.1836	-0.0000	-0.0000	0.0406	-0.0000
0.0000	-0.0000	-0.0000	0.1836	0.0000	-0.0000	0.0406
0.0000	0.0000	-0.0000	0.0000	0.0316	-0.0000	0.0000

Column 15

0.0000
 0.0000
 0.0000
 0.0000
 0.0000
 0.0000
 0.0000
 0.0000
 0.0000
 0.0000
 -0.0000
 0.0000
 0.0316
 -0.0000
 0.0000
 0.1807

e =

0.9910 + 0.0087i
 0.9910 - 0.0087i
 0.9672 + 0.0484i
 0.9672 - 0.0484i
 0.9341 + 0.1273i
 0.9341 - 0.1273i
 0.9341 + 0.1273i
 0.9341 - 0.1273i
 0.9417 + 0.0000i
 0.8971 + 0.0733i
 0.8971 - 0.0733i
 0.8971 + 0.0733i
 0.8971 - 0.0733i
 0.8604 + 0.0000i
 0.8604 + 0.0000i

$K_i =$

0.0000	-0.2520	0.3364
0.2520	-0.0000	0.3364
-0.0000	0.2520	0.3364
-0.2520	0.0000	0.3364

$K_s =$

Columns 1 through 7

0.0000	-6.2934	12.0470	0.0000	-3.1129	8.0631	11.6872
6.2934	-0.0000	12.0470	3.1129	-0.0000	8.0631	0.0000
-0.0000	6.2934	12.0470	-0.0000	3.1129	8.0631	-11.6872
-6.2934	-0.0000	12.0470	-3.1129	-0.0000	8.0631	0.0000

Columns 8 through 12

-0.0000	0.1567	2.7120	-0.0000	0.8989
11.6872	-0.1567	0.0000	2.7120	-0.8989
-0.0000	0.1567	-2.7120	0.0000	0.8989
-11.6872	-0.1567	0.0000	-2.7120	-0.8989

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